Technical Case Study: Achieving Zero-Void Encapsulation for High-Precision NEV Sensors

Abstract

In the rapidly evolving landscape of New Energy Vehicles (NEVs), sensor reliability is paramount. This case study explores how a leading automotive electronics manufacturer integrated the **HongJun HJ-VPM-001 Automatic Vacuum Potting Machine** to solve critical

insulation failures in pressure sensors, achieving an **85% reduction in production error rates** and a **100% void-free result**.

The Challenge

Atmospheric potting (potting under normal air pressure) often traps micro-bubbles (voids) within the epoxy or polyurethane resin. For high-precision NEV sensors:

1. **Dielectric Failure**: Voids lead to internal arcing and insulation breakdown.

2. **Mechanical Stress**: Trapped air expands/contracts with temperature, causing sensor drift.

3. **Moisture Ingress**: Voids provide pathways for humidity, leading to corrosion.

Customer Pain Point: Pre-automation, the client reported a 12% failure rate in high-voltage endurance tests.

The Solution: Vacuum-Integrated Dispensing

The HJ-VPM-001 system was deployed with the following technical specifications:

• **Vacuum Level**: < 500 Pa (residual pressure).

• **Dispensing Accuracy**: 0.01ml (Precision Screw Valve).

• **Process**: The sensor is placed in the vacuum chamber, air is evacuated, and the resin is dispensed directly into the housing while under vacuum.

Key Results

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The Accuracy Pivot: 0.01ml Precision

The integration of a **Precision Screw Valve** allowed the client to move away from traditional time-pressure valves. This ensured that even in high-vacuum environments—where resin viscosity can change—the volume dispensed remained constant at **0.01ml**, eliminating overflows and "starved" joints.

Conclusion

By pivoting from manual/atmospheric potting to the HJ-VPM-001 vacuum potting system, manufacturers can guarantee the integrity of electronic encapsulation, significantly reducing field failures and improving long-term product reliability.

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Metric	Before (Atmospheric)	After (Vacuum)	Improvement
Void Rate	15.4% (avg)	< 0.1%	99% Reduction
HV Test Failure	12.0%	1.8%	85% Reduction
Material Yield	88%	96%	9% Increase
Cycle Time	4.5 min/unit	2.1 min/unit	53% Faster

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